Engineering Mathematics - I · Unit_05 · MATLAB

DRAFT — UE26MA141A · Unit_05 · MATLAB · M-Files; Loop Control Statements; Conditional Statements

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

M-Files; Loop Control Statements; Conditional Statements

1 M-Files; Loop Control Statements; Conditional Statements

Study focus: M-Files; Loop Control Statements; Conditional Statements. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000042 · EV0000043

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I am Professor Nagesh Ham, faculty member, Department of Science and Humanities.

Key details

- This is the third session of MATLAB.
- In this session we learn M files.
- That is script files and function files, loop control statements, that is for loops and while loops.
- Finally condition statements.
- In MATLAB we write programs in M files.
- They are called M files because they must have dot M at the end of their name.
- The first one script files and then function files.
- Later we shall see what's the advantage of scries as well as function files.
- A script file is an M file with a set of valid MATLAB commands in it.
- Remember there are two ways to create a script file.
- The first one is this.
- On the command window we just write like this edit and then the file name which we want.
- You can see the output is going to be 600.
- We also find out the product of these three numbers PR O D A star B star C.
- Again, use this button run and you can see the answer is going to be this much.
- Again you give the command whatever you want.
- Let me find out a equal to 10 colon b equal to.

- Sign of y by two sine of pi by two then c equal to twenty right let me find out b equal to a plus b plus.
- Again go to this save.
- Again you can give.
- Let me give EC campus one.
- EC campus one dot m right now.
- the file has been created by the name ecampus1.m.
- You can see the answer is going to be 31.

Example or application from the lecture

- For example it should be like India.m then searching dot m and so on right so they are called m files because they must have dot m at the end of their file name there are two types of m files.
- Let's start with.
- Let us see how to create a script file.

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EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

Core explanation · 05:00–09:59

Assume you want to use this after one month of time.

Key details

- You want to compile this.
- Either you can choose the file from here from the current folder or you just type edit then the file name ecampus1.m again you can see the comments on the screen.
- function points A function is a group of statements that together perform a task.
- A function is a group of statements that together perform a task.
- Just clear the screen.
- It let me take such in.
- We run this, you can see answer is going to be 200.
- You have three inputs, A, B, C are the inputs, and the sum and product are the output.
- A B C are the input variables and sum and product are the output variables here.
- Can I cancel this.
- Is it clear?
- is clear then you can see the input variables are a comma b comma c a comma b comma c like this now remember the file name should be same as the one which we have used for creating the script file clear so this should be same as such in without M we should not use dot M here.
- Is it clear?
- Then we can use sum equal to A plus B plus C.
- Then Product PROD equal to A star B star C, B star C.
- Remember before we run this we have to say this.

- Remember again the file m should be same as the one what we have used for creating script file.
- We have saved a function file by searching.m.
- Remember in order to compile this we should use command window.
- search in of you can use whatever variables you want you can give okay I gave one comma one comma hundred right okay this is equal you can write it as s for sum p for product.

Example or application from the lecture

- So let us take an example.
- such in dot m let's create a script file for this searching now you give the command e equal to one B equal to two C equal to hundred then find out the sum equal to A plus B plus C then run this against answer is 103 then find out your product equal to a star b star c.
- So for that, what you need to do, you just clear the screen and remember for this particular example.

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EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 10:00–14:59

Equal to like this.

Key details

- You can see S is 102, the sum is 102, and the product is 100.
- See if you want to.
- Add some of the three numbers.
- What we should do?
- We have to manually update a script file every time.
- Just copy paste here.
- You can change the number here.
- Then hundred, then thousand.
- Use we can see the output.
- It means instead of manually updating the script each time, we can make program more flexible by converting it to a function.
- Is it clear?
- Clear the screen.
- Next we shall move on to low control statements.
- Using loop control statements, we can repeatedly execute a block of code.
- There are two types of loops.
- One is for loops and while loops.
- For loop is used to repeat a statement or a group of statements for a specific number of times.
- For loop is used to repeat a statement or a group of statements for a specific number of times.
- Remember here what we see DISP of A, this is not a display command.

- It displays the values of A without a variable name.
- Is it clear?
- You can see the this is A is a variable whose values are 1, 2, 3, 4.
- Displays the values of A without printing a variable name that is A.
- Here m start from 1 goes up to 3 so you can write number equal to 1 divided by m plus 1 then n.
- When m is two number is one zero point double three double three and so on.
- Is it clear?
- Is it clear?
- While loop repeatedly execute program statements as long as the expression remains true.
- Then now the while loop starts.
- And the condition here is A is less than fifteen.

Example or application from the lecture

- Now for example, for the same command, we can give some number like this.
- Let's talk about for loops.
- For example, we consider this.

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EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

Core explanation · 15:00–19:59

the while will run till the value of A is less than 15.

Key details

- Next we print the value F P R I N D print f printf value of e colon percentage D slash N.
- It means it prints a current value of A and then A equal to A plus 1 is executed.
- It means this is the output for the following command.
- It means we first define a equals to 10.
- We define the variable a equal to 10.
- Now the while loop starts and the condition for a is a is less than 15.
- It means that the while loop will run till the value of A is less than 15.
- Currently the value of A is 10.
- The line A equal to A plus 1 add 1 to A and then while conduction is checked.
- That's why the output is like this 10 to 14.
- And these are the some of the examples what we have discussed.
- There will be some situation where a program or a particular block has to be executed only when a specific condition is true.
- The condition statements will be useful in such situations.
- The following are the condition statements that we can use in MATLAB.
- If and if else and if else if else if else and switch case.
- So these are the some of the condition statements what we use in MATLAB.
- If and statement.
- Number equal to 25.

- Here you can see that number is 25.
- So this 25 is more than 20, so it is true, therefore it prints the output like this.
- Because the condition is true here.
- Number is 25.
- Here the condition is false, that's why it displays nothing on the screen.
- So next we have if else and statement.
- An if and statement can be followed by an optional L statement which executes when the expression is false.
- If the number is less than 20, so this is false, therefore it prints the second line.
- The number is not less than 20.
- The number is not less than 20.
- So if this condition is not true, therefore it executes the next line.
- Next if else if else if else and so this is a chain.
- So next we have a switch statement.
- The switch statement is used to test for equality against a set of known values.

Example or application from the lecture

- After understanding loop control statements, so let's move on to conditional statements.
- Let us model the loop a condition statements.
- It decides whether a particular block of code has to be executed or not based on the given Boolean condition that is 0 or 1.

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EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

Core explanation · 20:00–24:59

This one when we run the file it displays the following, is it?

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EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
M-Files; Loop Control Statements; Conditional Statements transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/003_M-Files_Loop_Control_Statements_Conditional_Statements__d8fc5a0a5b8f6dbe.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
M-Files; Loop Control Statements; Conditional Statements transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/003_M-Files_Loop_Control_Statements_Conditional_Statements__d8fc5a0a5b8f6dbe.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
M-Files; Loop Control Statements; Conditional Statements transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/003_M-Files_Loop_Control_Statements_Conditional_Statements__d8fc5a0a5b8f6dbe.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9986; only the JSON segmentation supplies the displayed timestamps.
M-Files; Loop Control Statements; Conditional Statements transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/003_M-Files_Loop_Control_Statements_Conditional_Statements__d8fc5a0a5b8f6dbe.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 4 rendered pages in 001_Slides__7ac00f797de85ee4.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 002_Slides__ef919841425356a4.slides.md have no readable `.slides.md` text; their original page images are included.

3. 1 rendered pages in 003_Slides__3d7af4a88d5b4490.slides.md have no readable `.slides.md` text; their original page images are included.
4. 1 rendered pages in 004_Slides__4150cd4c33be9a2c.slides.md have no readable `.slides.md` text; their original page images are included.
5. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
6. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
7. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
8. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
9. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
10. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
11. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00010	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0034. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0035 to s0051. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00011	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0034. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0035 to s0051. Sidecar variants compare...
SRC00012	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0034. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0035 to s0051. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00013	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0034. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0035 to s0051. Sidecar variants compare...

C Revision and retrieval practice